

1. Introduction to AI & Data Analysis

This lab introduces the fundamentals of Artificial Intelligence, the importance of data in research, common data types, the role of AI in data analysis, and a basic hands-on demonstration using Google Colab with Gemini.

1. 1. Overview of AI

Before exploring advanced analytical techniques, it is essential to distinguish the foundational pillars that shape the field:

Artificial Intelligence (AI):

A broad domain of computer science focused on enabling machines to perform tasks that typically require human cognitive abilities, such as reasoning, problem-solving, pattern recognition, and decision-making.

Machine Learning (ML):

A specialized subset of AI that uses data and algorithms to enable systems to learn from experience. ML models improve their performance over time without requiring explicit programming for each individual task.

Data Analysis:

A systematic process that involves inspecting, cleaning, transforming, and modeling data to uncover meaningful patterns, generate insights, and support evidence-based decisions.

The Integration:

In modern research environments, AI-driven data analysis combines intelligent algorithms with analytical workflows, enabling the processing of large, complex, or unstructured datasets that traditional statistical approaches cannot efficiently handle.

1. 2. Data in research and Types of data

Why Data Types Matter?

- Data's DNA: Before analyzing data, a manager must understand its "DNA."
- Proper Analysis: Data types dictate which statistical tests are valid and which visualizations are appropriate.
- Garbage In, Garbage Out: Incorrectly identifying a data type leads to flawed models.
- Storage & Cost: Different types require different storage architectures (SQL vs. NoSQL).

Quantitative (Numerical) Data

- Quantitative data represents measurable quantities.
- It is further divided based on the "fineness" of the measurement.

A. Discrete Data

B. Continuous Data

Discrete Data: the countable data

- Definition: Countable values with no "in-between" (integers).
- Discrete data represents values you can count one by one.
- No decimals or fractions are allowed.
- Usually used for counting people, objects, or events.
- Example: the number of students in a class (30, 31, 32) or how many books you have (5, 6, 7)

Discrete Data – Analytic Tips

- Visualized with Bar Charts because values are separate and countable.
- Frequency tables work well for summarizing how frequent each value occurs.
- Often analyzed using counts, proportions, or percentages.
- Good for comparing categories or distinct groups.
- Example analyses: number of customers per day, number of items sold in each category

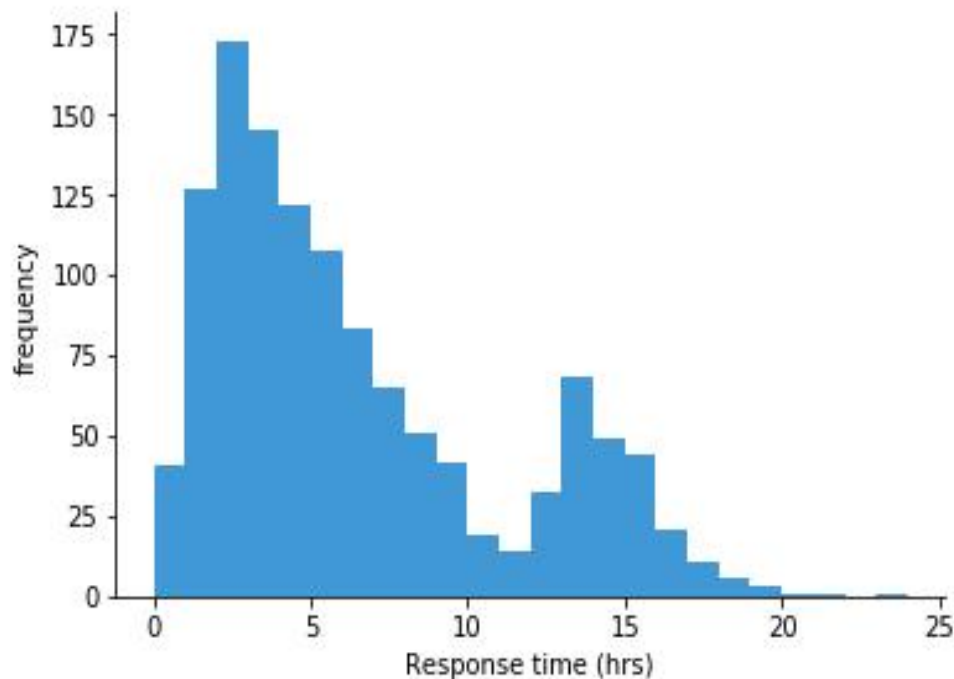
Continuous Data: Measurement

- Continuous data represents values you can measure.
- These values can take any value within a range, including decimals.
- Can be infinitely precise depending on the measuring tool.
- Usually used for measuring time, amount, distance, or size.
- Example: someone's height (150.5 cm), or the temperature (29.8°C)

Continuous Data – Analytic Tips

1. Visualized with Histograms or Line Charts - the data flows across a range.
2. Descriptive statistics (mean, median, standard deviation) are applied.
3. Useful for understanding trends, patterns, and distributions.
4. Often used in forecasting and modeling - the values change smoothly.
5. Example analyses: sales revenue over time, temperature trends, processing time in minutes.

Histogram for Continuous data



Qualitative (Categorical) Data

- Qualitative data refers to information that describes qualities, characteristics, or categories rather than numbers.
 - They help us understand what type or which group something belongs to.
 - They are not measured with numbers; instead, they represent labels or categories used for classification in business.
- i. A. Nominal Data
 - ii. B. Ordinal Data

Nominal Data (Labels Without Order)

1. Nominal data represents names or labels with no ranking or order.
2. Categories are different, but none is “higher” or “lower.”
3. Common in business for identifying customer groups, product categories, or market segments.
4. Examples: customer gender, product brand, payment method, industry type.

Nominal Data - Analytic Tips

- Usually visualized with Bar Charts or Pie Charts or Heatmaps because each category is separate.
- Best summarized using frequencies and percentages.
- Works well for customer segmentation, brand comparison, and preference analysis.

Heatmaps in Nominal data



Source: raw.githubusercontent.com

Ordinal Data: Ordered Category Data

- Ordinal data uses categories, but these categories have a meaningful order.
- The order matters (e.g., “High” is above “Medium”), but the gaps between levels are not equal.
- Common in business for ratings, satisfactions, rankings, and priority levels.
- Examples: satisfaction levels (Poor → Fair → Good → Excellent), education level, service priority.

Ordinal Data - Analytic Tips

- Usually visualized with Ordered Bar Charts to show ranking clearly.
- Best described using the median or mode, not the mean.
- Useful for analyzing customer satisfaction, service quality, and preference intensity.

Unusual Data Structure

- In business and data analytics, data comes in different forms and formats, and understanding these types is crucial for making effective decisions.
- Broadly, data can be quantitative or qualitative, and for practical analysis, it is often classified as unusual structures.
- Structured, Semi-structured, and Unstructured
- This classification helps us determine how the data can be stored, accessed, and analyzed.

Structured Data

- Structured data is information that is organized, searchable, and stored in fixed formats such as tables, rows, and columns.
- It fits neatly into databases and spreadsheets, making it easy for analysts to query and run calculations.
- In business, this includes data like sales numbers, customer IDs, transaction records, inventory lists, and performance metrics.

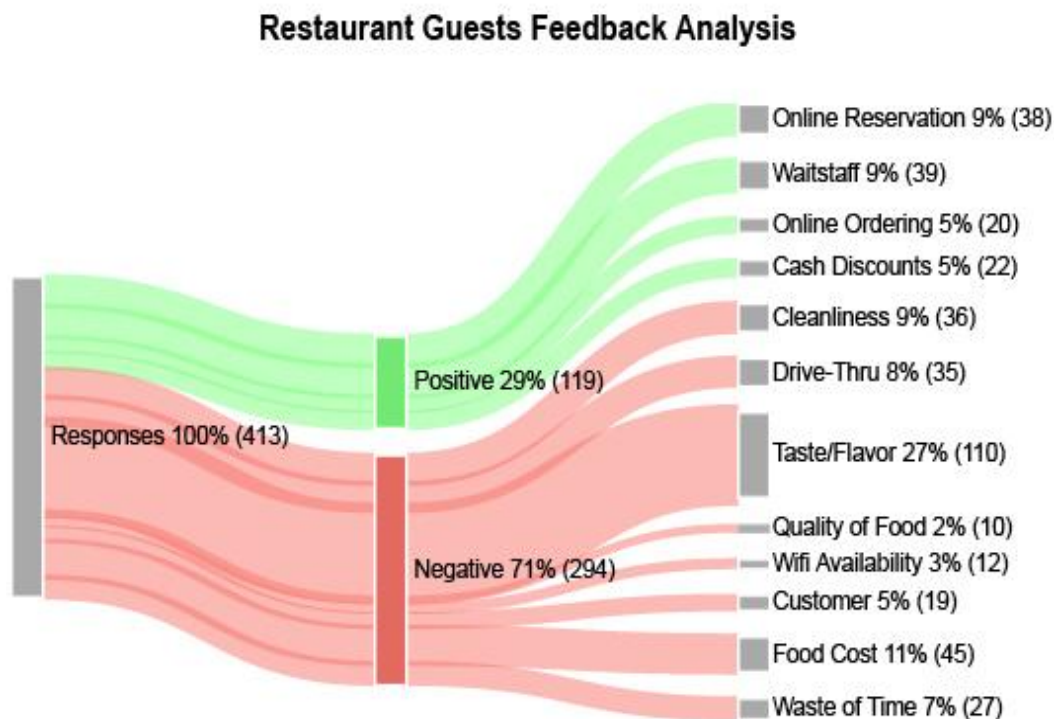
Structured Data - Analytic Tips

- Best analyzed using SQL queries, dashboards, and statistical models.
- Works well with BI tools because it is clean, consistent, and machine-readable.
- Easy to visualize using tables, bar charts, line charts, and KPIs dashboard.

Unstructured Data

- Unstructured data has no fixed format, making it harder to store, search, and analyze directly.
 - This includes text, images, audio, video, emails, customer reviews, chat messages, and social media posts.
 - Businesses use unstructured data to understand customer behavior, sentiment, and emerging patterns that are not captured in numeric tables.
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- Unstructured Data - Analytic Tips
 - Requires advanced analytics such as natural language processing (NLP), text mining, image recognition, or AI-based tools.
 - Useful for discovering customer emotions, trends, opinions, and hidden insights.
 - Best visualized through word clouds, sentiment charts, tag clusters, or dashboards combining text and numbers.

Sentiment Charts in Unstructured data



1. 3. Role of AI

The AI-Driven Research Workflow

Traditional research is often linear and manual. AI transforms this into an iterative, automated loop to ensure data has been perfectly analyzed.

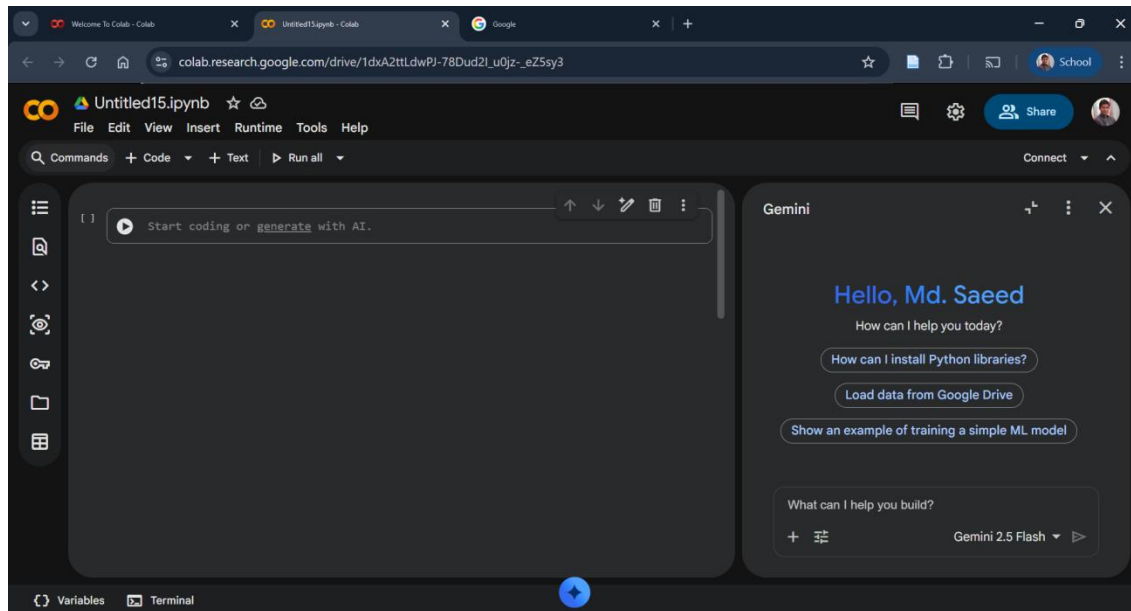
The Workflow Diagram

Raw Data → AI Pre-processing → Pattern Recognition → Predictive Modeling → Insights/Action

1. **Ingestion:** Scrapping data from sensors, journals, or databases.
2. **Cleaning (Auto-Wrangling):** AI identifies missing values or "noise" (errors) and fixes them.
3. **Analysis:** The model finds correlations (e.g., "Every time Temperature X rises, Yield Y drops").
4. **Validation:** Comparing AI findings against "Ground Truth" (known facts).

1. 4. Demo of Colab + Gemini

Open Google Colab and login there.



There are two types of cells in the notebook (1) Code Cell and another is (2) Text Cell

